2024 SSMO Accuracy Round

SMO Team

AR 1: Let a time of day be three-full if exactly three of its digits are 3s when displayed on a 12-hour clock in the *hh*:*mm*:*ss* format. How many seconds of the day are three-full?

AR 2: Equilateral triangle N is inscribed within circle O. A smaller equilateral triangle P is inscribed within N such that the vertices of P lie on the midpoints of the sides of N. The ratio of the area of O to the area of P can be expressed as $\frac{a\pi\sqrt{b}}{c}$, where a, b, and c are positive integers, a and c are relatively prime, and b is not divisible by the square of any prime. Find $a+b+c$.

AR 3: Three distinct integers a, b, and c with $1 \leq a, b, c \leq 10$ are chosen uniformly at random. The probability that $5 \mid a! + b! + c!$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

AR 4: Right triangle ABC has a right angle at C and hypotenuse 1. Let points D and E lie on line AC such that $\angle BDC = \angle BEC = 45^\circ$ and points A, D, C, and E lie in that order. Given that $EA = 13DA$, the area of $\triangle ABC$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

AR 5: Let $ABCD$ be a convex quadrilateral such that $\angle ABC = 120^\circ$ and $\angle ADC = 60^\circ$. If $AB = BC = CD = 5$, the area of $ABCD$ can be expressed as $\frac{a\sqrt{b}}{c}$, where a, b, and c are positive integers, a and c are relatively prime, and b is not divisible by the square of any prime. Find $a+b+c$.

AR 6: Three six-sided dice are rolled. Then, the product of the three numbers on the top faces is calculated. The probability that the product is not a perfect square can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

AR 7: Let a, b, c, and d be positive real numbers such that

$$\begin{aligned} a^2 + d^2 &= b^2 + c^2 = 425, \\ ac + bd &= 375, \text{ and} \\ ab + cd &= 408. \end{aligned}$$

Find the value of $a+b+c+d$.

AR 8: $ABCD$ is a convex cyclic quadrilateral with $AB = 2$, $BC = 5$, $CD = 10$, and $AD = 11$. Let W, X, Y, and Z be the midpoints of sides AB, CD, BC, and DA, respectively. Find $|WX^2 - YZ^2|$.

AR 9: In the game of S-set, there are 3^{12} distinct cards. Each card has 12 traits, each trait is one of three variants A, B, and C, and no two cards have the same variants for all 12 traits. A full house in the game of S-set is defined to be a nonempty hand of cards in which for each trait, all cards in the hand either share the same variant or have all different variants. The number of hands that are full houses can be expressed as $2 \cdot 3^a + 3^b$, where a and b are positive integers. Find $a + b$.

AR 10: Bobby is spinning a rigged wheel with three sections labeled A, B, and C. Two integers $0 \leq x, y \leq 50$ are chosen independently and uniformly at random, such that there is an $x\%$ chance the wheel lands on A and a $y\%$ chance the wheel lands on B. Given that the wheel lands on C the first time, the probability that it will land on C if Bobby spins it again can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.